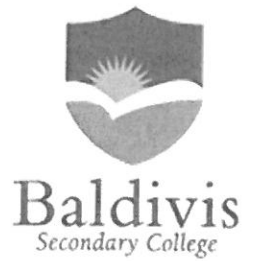


Year 10
Test 3
Measurement
2019



Name: _____ Class: _____ Time: ~~60~~ minutes Total Mark: 57 / 38

Show working and answers on this sheet. Show working in sufficient detail to support your answers. Incorrect answers without supporting reasoning will be allocated zero marks.

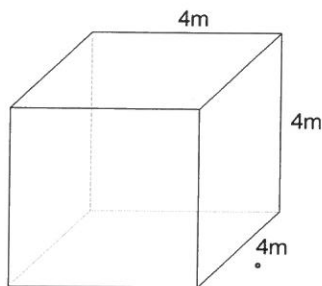
Resources: 1 A4 page of notes (2 sided), Yr 10 formula sheet, Calculator.

Surface Area	19
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Question 1	2, 3, -5 marks
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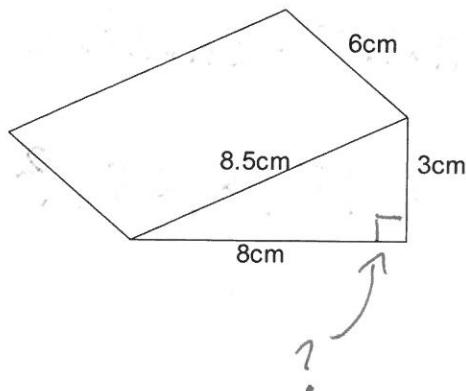
Determine the **surface area** of the following 3D Shapes:

a)



$$\begin{aligned} (4 \times 4) \times 6 & \checkmark \\ 16 \times 6 & \\ = 96 \text{ m}^2 & \checkmark \end{aligned}$$

b)



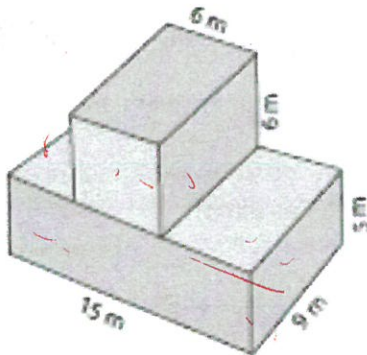
$$\begin{aligned} 6 \times 8.5 &= 51 + \\ \frac{1}{2} \times 3 \times 8 \times 2 &= 24 + \\ 3 \times 6 &= 18 + \\ 8 \times 6 &= 48 + \\ \text{TSA} &= \underline{141 \text{ cm}^2} \checkmark \end{aligned}$$

Question 2

8 marks

Calculate the surface area of the following composite solids

a)



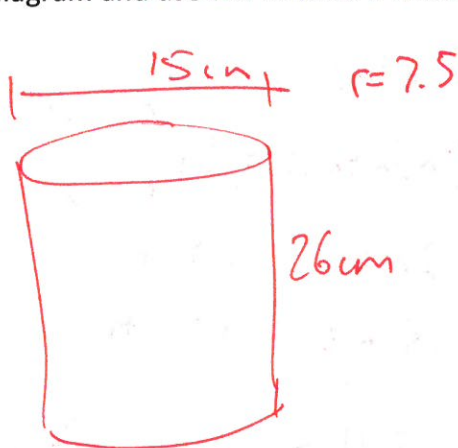
$$\begin{aligned}
 \text{Top/bottom: } 9 \times 15 \times 2 &= 270 \\
 \text{sides: } (5+6) \times 9 \times 2 &= 198 \\
 \text{F/B: } (6 \times 6) + (15 \times 5) \times 2 &= 222 \\
 \text{TSA} &= \underline{690 \text{ m}^2}
 \end{aligned}$$

Question 3

3 marks

Calculate the surface area of a cylinder with diameter of 15 cm and a height of 26 cm.
(HINT: Draw a diagram and use the formula if needed)

(closed ends)

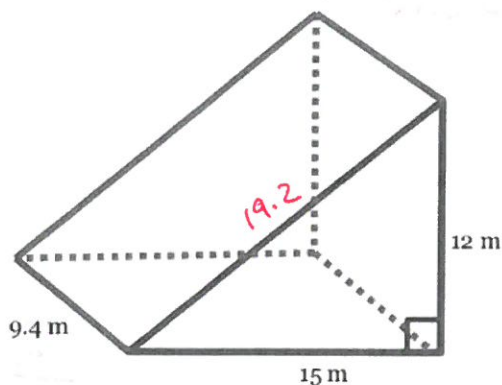


$$\begin{aligned}
 2\pi \times 7.5^2 + 2\pi \times 7.5 \times 26 \\
 = 1578.65 \text{ cm}^2 \checkmark
 \end{aligned}$$

Question 4

3 marks

Calculate the surface area of the following triangular prism



$$12^2 + 15^2 = c^2$$

$$c = 19.2 \checkmark$$

$$\frac{1}{2} \times 15 \times 12 \times 2 = 180$$

$$9.4 \times 19.2 = 180.48 \checkmark$$

$$15 \times 9.4 = 141 \checkmark$$

$$12 \times 9.4 = 112.8$$

$$TSA = 614.3 \text{ m}^2 \checkmark$$

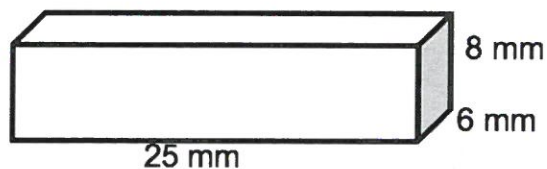
Volume 9

Question 5

3 marks

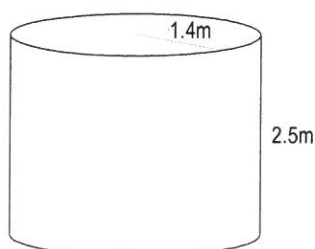
Calculate the Volume of the following 3D Shapes.

a)



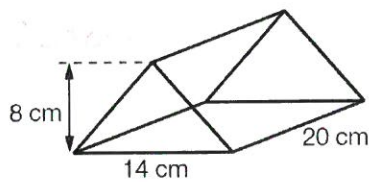
$$8 \times 6 \times 25 = 1200 \text{ mm}^3$$

b)



$$\pi \times 1.4^2 \times 2.5 = 15.4 \text{ m}^3$$

c)

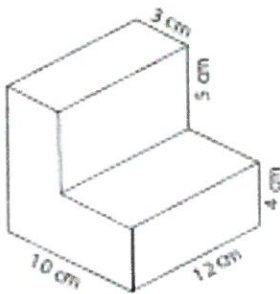


$$\frac{1}{2} \times 14 \times 8 \times 20 = 1120 \text{ cm}^3 \checkmark$$

Question 6

4 marks

Calculate the volume of the following solid



$$(10 \times 4 + 5 \times 3) \times 12$$

$$55$$

$$55 \times 12 \checkmark$$

$$= 660 \text{ cm}^3 \checkmark$$

3
?
too many

Question 7

2 marks

A cylinder has a volume of 375 cm^3 and a height of 15 cm , find its radius.

$$375 = \pi r^2 \times 15 \checkmark$$

$$r^2 = 7.96$$

$$r = 2.8 \text{ cm} \checkmark$$

END OF TEST